



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Team Cognition

Team Leader Name : Saurav Gupta

Problem Statement : **Air-Gapped Predictive Copilot for Secure MPLS Operations**

Team Members

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Opportunity should be able to explain the following:

- How different is it from any of the other existing ideas?

- 1) **Zero Cloud Dependency:** While most AI diagnostic tools rely on external APIs (like OpenAI), our solution runs a localized, 7-billion parameter LLM (Qwen2) entirely offline on local GPU clusters (via SLURM).
- 2) **Beyond Basic Dashboards:** Traditional NOC tools (like SolarWinds) only provide alerts and charts, leaving humans to find the fix. Our system autonomously reads the raw router logs and writes the fix for you.
- 3) **Context-Aware via RAG:** It doesn't just guess. It uses a local Chroma Vector Database to cross-reference real-time anomalies with actual, offline enterprise networking runbooks before generating an answer.

Opportunity should be able to explain the following:

- How will it be able to solve the problem?

- 1) **Eliminates Manual Log Hunting:** During critical network failures (like BGP Route Flaps), operators waste precious minutes manually searching through Syslog data and PDF manuals.
- 2) **Instant Root Cause Analysis (RCA):** The Copilot instantly ingests the failing telemetry, analyzes the logs, and streams an automated diagnosis in real-time.
- 3) **Actionable Remediation:** Instead of just telling the operator that a tunnel is down, the AI provides the exact Cisco/Juniper CLI commands needed to safely repair the issue without causing cascading failures.

Opportunity should be able to explain the following:

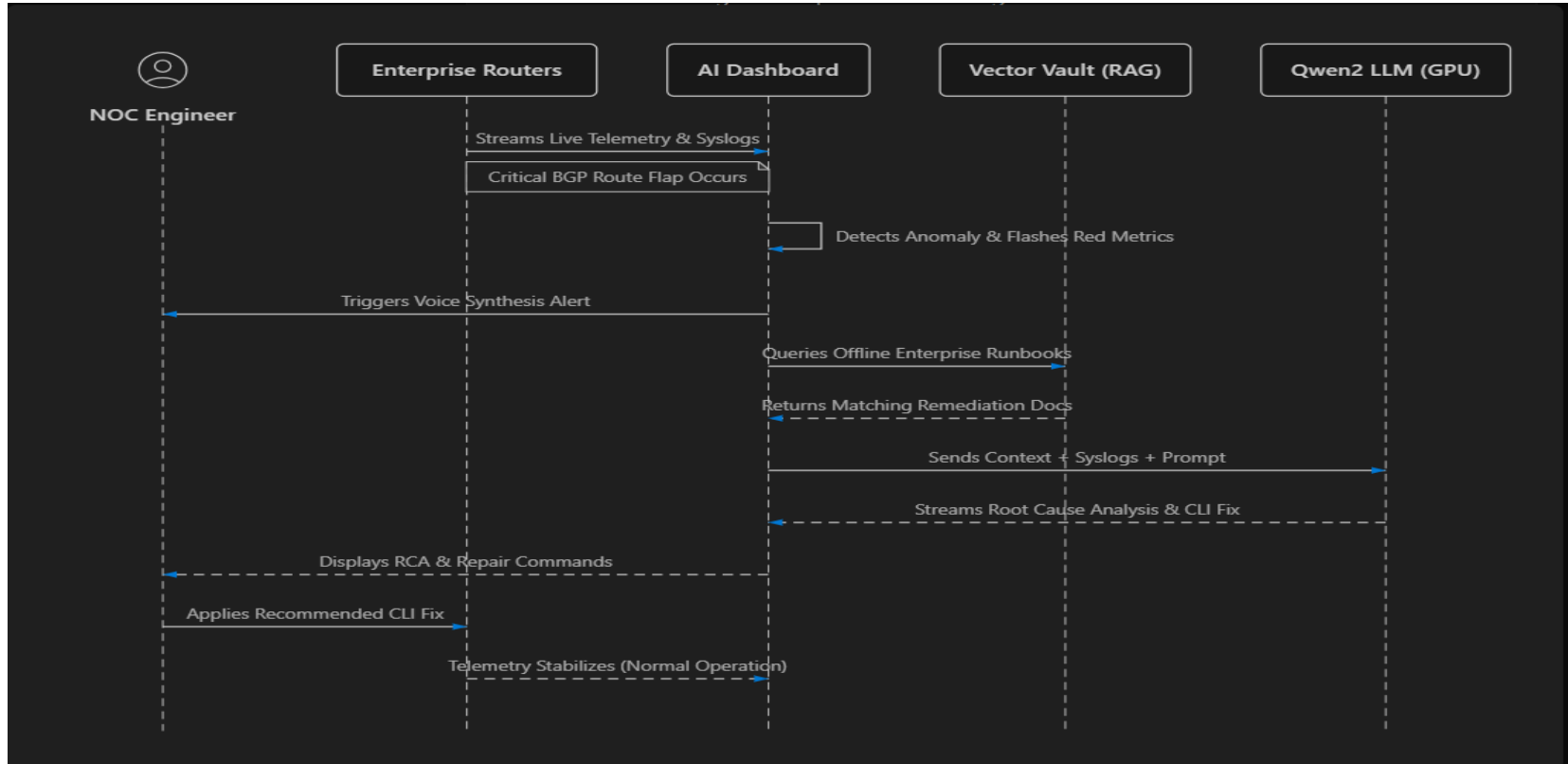
- USP of the proposed solution

- 1) **100% Air-Gapped Data Privacy:** Guaranteed zero data exfiltration. Highly sensitive ISRO network telemetry never leaves the secure local environment.
- 2) **Autonomous RAG Architecture:** Combines the reasoning power of an advanced LLM with a dedicated offline vector vault of operational runbooks for highly accurate, hallucination-free diagnostics.
- 3) **Multimodal Alerting:** Features a custom-built, glassmorphic dashboard with live telemetry integration and text-to-speech voice synthesis for instant, hands-free operator warnings

List of features offered by the solution

- a) **Air-Gapped AI Engine:** Powered by a localized 7-billion parameter LLM (Qwen2) running entirely offline on high-performance GPUs (via PyTorch) to ensure absolute data sovereignty.
- b) **Autonomous Root Cause Analysis (RCA):** Automatically parses raw router syslogs and network anomalies to stream real-time diagnostic reports with precise severity classifications.
- c) **RAG-Powered Telemetry Vault:** Utilizes an integrated ChromaDB vector database to cross-reference live network failures against offline, secure enterprise runbooks.
- d) **Actionable CLI Remediation:** Doesn't just flag errors—it generates the exact, step-by-step mitigation commands needed to restore network stability.
- e) **Interactive Telemetry Dashboard:** A premium, glassmorphic UI that visualizes live system stress metrics (Latency, Packet Loss, BGP Peers) alongside the Copilot chat.
- f) **Text-to-Speech Voice Alerts:** Built-in audio synthesis that instantly vocalizes critical network failures for hands-free NOC operators.

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution


AI NOC Copilot
ENTERPRISE

AUTONOMOUS SD-WAN DIAGNOSTICS

Voice: ON

Architecture

Scenarios

SECURE

NORMAL OPERATION

SYSTEM STATUS

PREDICTED FAILURE

AVG LATENCY	PACKET LOSS	ACTIVE TUNNELS	BGP PEERS	UPTIME	THREAT LEVEL
207.8 ms	4.67%	28	5	99.97%	CRITICAL

[8:11:50 PM] WARNING: Interface GigE0/1 utilization at 76%

[8:11:50 PM] INFO: MPLS label switch path verified OK

[8:11:44 PM] INFO: BGP peer 10.0.0.1 session ESTABLISHED

[8:11:38 PM] CRITICAL: MPLS LSP down: PE1-to-PE3 tunnel unreachable

A branch office BGP session is flapping every 30 seconds with repeated NOTIFICATION messages. The router log shows: %BGP-3 NOTIFICATION: received from neighbor 10.0.0.1 (hold time expired). Diagnose the root cause and recommend remediation steps.

[SEVERITY: WARNING]

Diagnosis:
 The issue described indicates that there's a BGP session between your router and the neighbor at IP address 10.0.0.1 experiencing frequent hold time expiration events, leading to repeated NOTIFICATION messages. This behavior typically suggests one or more of the following issues:

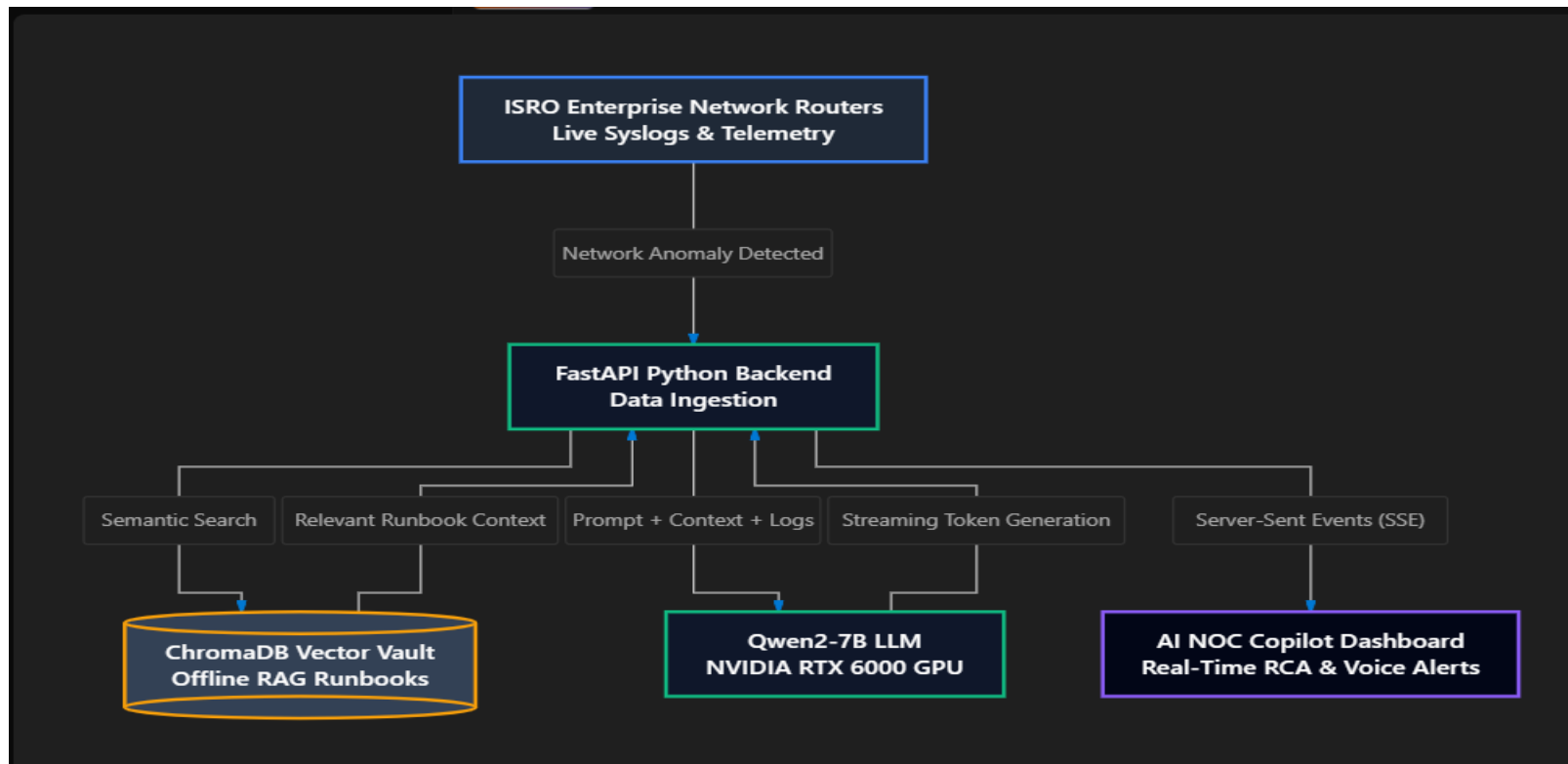
- Misconfiguration:** Incorrect BGP configuration settings, such as incorrect AS numbers, incorrect holdtime/keepalive intervals, or misconfigured routing policies.
- Network Connectivity Issues:** Fluctuations in network connectivity between the two routers could lead to hold-time expiration, especially if the link quality is poor or unstable.
- Routing Protocol Issues:** Potential issues within the BGP protocol itself, such as incorrect route advertisement or withdrawal mechanisms.
- Resource Limitations:** Insufficient resources on either end of the BGP session might cause the hold time to expire prematurely.

Recommendations for Remediation:
Step 1: Verify Configuration

- Check BGP Configuration:** Ensure that both routers have the correct AS number, holdtime, and keepalive intervals configured. The holdtime should be twice the keepalive interval. For example, if the

Paste router logs or diagnostic queries...

Architecture diagram of the proposed solution



Technologies to be used in the solution:

1. **Core Large Language Model:** Qwen2-7B (Open-source, state-of-the-art reasoning model).
2. **AI & Machine Learning Framework:** PyTorch and HuggingFace transformers (optimized for NVIDIA GPU acceleration).
3. **Vector Database (RAG Vault):** ChromaDB (for secure, offline semantic search of enterprise runbooks and telemetry history).
4. **Backend API Server:** Python, FastAPI, and Uvicorn (handling asynchronous Server-Sent Events for real-time text streaming).
5. **Frontend Dashboard:** HTML5, Vanilla JavaScript, and CSS3 (creating a lightweight, dependency-free interface with glassmorphic design and Web Speech API for voice synthesis).
6. **Infrastructure:** SLURM Workload Manager (for HPC cluster deployment and GPU allocation).

Estimated implementation cost (optional):

Since our solution is designed to be air-gapped and relies entirely on open-source software, the ongoing operational costs are exceptionally low compared to enterprise SaaS alternatives.

- 1. Software Licensing Cost:** ₹0 (Built entirely on open-source frameworks like PyTorch, FastAPI, and ChromaDB. No costly licenses like SolarWinds).
- 2. Cloud API Cost:** ₹0 (Because the LLM runs 100% locally, there are zero recurring API fees to companies like OpenAI or AWS).
- 3. Hardware Cost:** The only requirement is an enterprise GPU (e.g., NVIDIA RTX 6000 or A100). If ISRO already has an HPC cluster, this cost is ₹0 (absorbed by existing infrastructure). If purchasing a dedicated AI workstation, a one-time Capital Expenditure of approx. ₹4,00,000 to ₹6,00,000 is required.
- 4. Return on Investment (ROI):** Extremely high. By automating Root Cause Analysis and reducing Network Downtime (MTTR) by minutes or hours per incident, the system pays for its hardware cost in the first major outage prevented.

Implementation Video Link:

[Drive Video Link](#)

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THANK YOU